

# **CONTRIBUTION OF ENGINEERING IN BIOLOGICAL DOMAIN**

- Biomedical engineering applications include the development of biocompatible prostheses, various diagnostic and therapeutic medical devices ranging from clinical equipment to micro-implants, imaging technologies such as MRI and EKG/ECG, regenerative tissue growth, and the development of pharmaceutical drugs .

# Microscope

- The microscope is a device that magnifies objects or organisms that are too small to see with the naked eye.
- The invention of the microscope allowed scientists to see cells, bacteria, and many other structures that are too small to be seen with the unaided eye.
- It gave them a direct view into the unseen world of the extremely tiny.

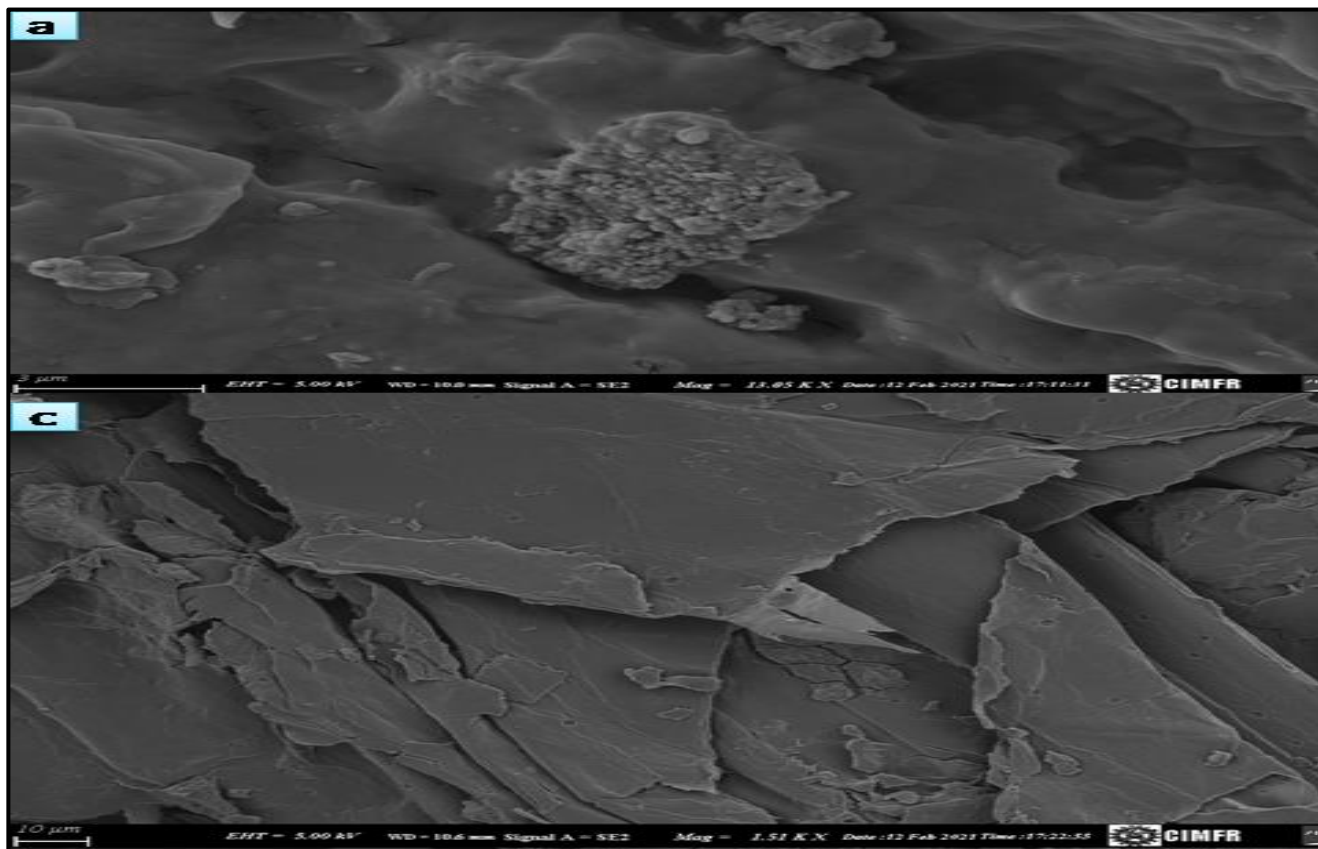


## Importance of microscope in our daily life

- By using Microscopes scientists, researchers and students were able to discover the existence of microorganisms, study the structure of cells and see the smallest parts of plants, animals and fungi.
- Microscopes are not just used to observe cells and their structure but are also used in many industries.
- For example, electron microscopes help create and observe extremely tiny electrical circuits found on Silicon microchips. Scanning microscopes are much more sophisticated and they have higher magnifications than light-refracting microscopes.
- Microscopes are also used in the field of genetics.
- Microscopes are also used to diagnose illness in hospitals and clinics all over the world. Microscopes magnify the blood samples, so the doctors or the pathologist can see the viruses and the parasites attacking the red blood cells and take the necessary steps to cure it.

- Microscopes use the simple visible light refracting lenses. Electrons, x-rays and infrared rays are able to resolve the viruses which are far smaller than any cell. They enlarge the view of tiny viruses, which allows scientists to develop the vaccines and cures for infectious diseases in the humans and the animals.
- Scanning electron microscopes (SEM) have magnifications up to several million times to view the molecules, the viruses and the nano-particles.
- SEM was used for surface analysis of adsorbent.

## Scanning electron microscopes (SEM)



**Fig: FE-SEM image of PANI@SB (a) before & (c) after Cr (VI) adsorption**

# Imaging Techniques



## Imaging Techniques

- Medical imaging or **Imaging Techniques** is the process of using specialized techniques (such as ultrasound, tomography, or radiography) to produce an image of internal organ of the body that is not visible to the naked eye.
- Biomedical imaging techniques had significantly improved the health care of patients.
- It reduced the high risk of human errors with improved accuracy in disease detection and surgical procedures.
- Advanced imaging techniques play a critical role across healthcare, with a hand in diagnostics and disease monitoring.

## Imaging Techniques

- A beam of wave passes through the body/area under diagnosis, transmits or reflects back the radiation which will be captured by a detector and processed to get an image pattern.
- CT involves the use of X-rays, whereas radio frequency waves and gamma rays are used for MRI.
- Radiography uses electromagnetic radiation to take images of the inside of the body. The most well-known and common form of radiography is X-ray.

## Classification of Imaging Techniques

The main types of imaging used in modern medicine are:

- X-rays (radiographs)
- Ultrasound
- Computed tomography (CT Scan)
- Magnetic resonant imaging (MRI)

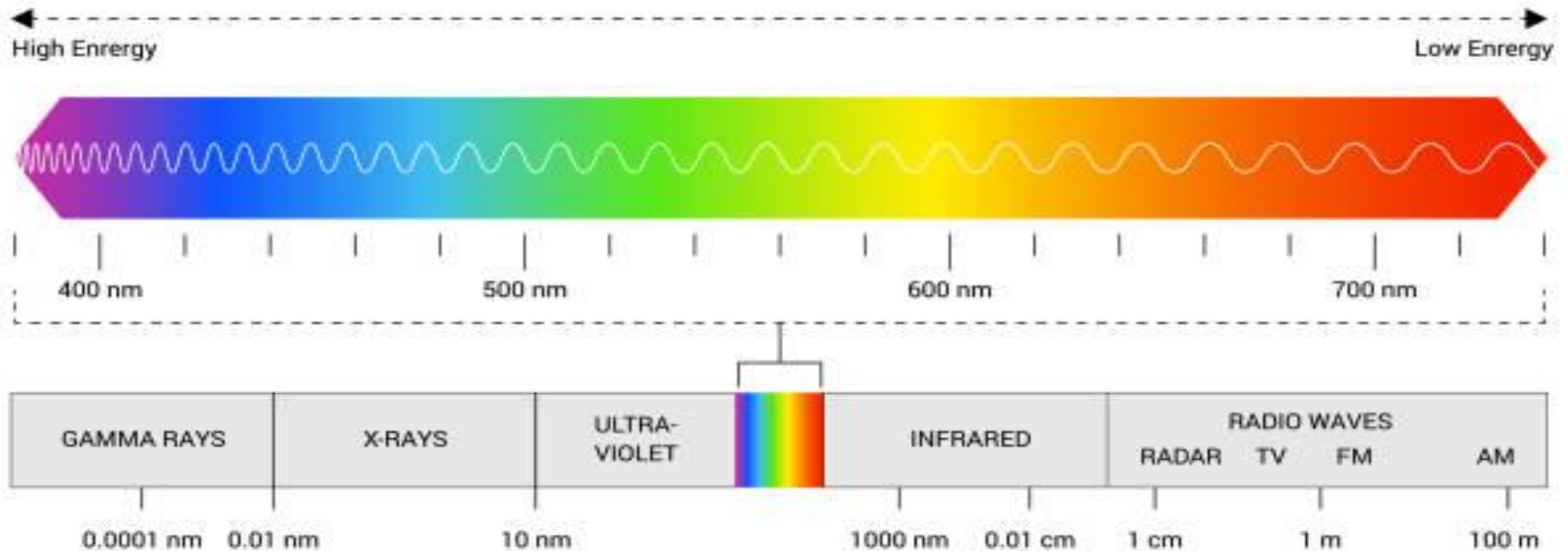
# X-rays

- X-rays are the most common diagnostic imaging technique.
- It is a form of high-energy electromagnetic radiation that passes through most objects.
- During an X-ray test, X-rays travel through the body and are absorbed in different quantities by different organs.
- For example, the calcium content in bones absorb x-rays better than other tissue, such as fat.
- An X-ray detector receives the X-rays after they pass through the body, generating an image that appears in shades of gray.
- X-ray effortlessly pass through skin, bone and metal to produce images that the human eye would never be able to see.

- X-ray imaging is often the first imaging technique used when a patient has a problem.
- They are used for a wide variety of purposes, ranging from looking for dental cavities to detecting cancer.
- X-rays are also used in other imaging techniques, like CT scans.



# VISIBLE SPECTRUM



## Application of X-ray

- **Airport security:** X-ray machines are used in airports to check luggage etc. In the past few years full body x-ray scans have also emerged as an additional security measure.
- **Industry:** X-rays are used in industry to inspect product made by various kinds of materials.
- **Science:** In Science X-rays are used to analyze the arrangement of atoms in many kind of substances, particularly crystals.
- **Consumer goods :** X-rays are also used in consumer goods the manufactures treat certain kinds of plastic to check the quality of many mass produced products.

- **Research:** Used in research involving quantum mechanics theory and crystallography (examine arrangement of atoms in solid).
- X-rays are used in medicine for medical analysis. Dentists use them to find complications, cavities and impacted teeth.
- Detects bone fractures, certain tumors and other abnormal masses, pneumonia, some types of injuries, calcifications, foreign objects, or dental problems.
- X-ray can be used to identify lung diseases such as pneumonia, lung cancer or pulmonary oedema.
- Compared to other advanced imaging modalities like [CT scans](#) or [MRIs](#), X-rays are a relatively affordable tool.

# Ultrasound

- Ultrasound (US) scans use sound waves to produce images of internal structures, most commonly abdominal and pelvic organs, the heart, blood vessels, and developing fetuses in utero.
- In scanning with ultrasound, high-frequency sound waves are transmitted to the area of interest (body part) with a device called a transducer and the returning echoes reflected by internal structures are recorded.
- The transducer is placed directly on top of the skin, which has a gel applied to the surface.
- These echoes return to the transducer and are transmitted electrically onto a viewing monitor.

# What is an Ultrasound Test?

- The echo images are then recorded on a plane film and can also be recorded on videotape.
- After the ultrasound, the gel is easily wiped off.
- The technical term for ultrasound testing and recording is "sonography."
- Ultrasound testing is painless and harmless.
- Ultrasound tests involve no radiation and studies have not revealed any adverse effects.



## Application of Ultrasound

- Ultrasound has been used in a variety of clinical settings, including obstetrics and gynecology, cardiology and cancer detection.
- The main advantage of ultrasound is that **certain structures can be observed without using radiation.**
- **It can also be done much faster than X-rays or other radiographic techniques.**
- It can be used to **examine internal structures of the abdomen.**
- It can detect fluid, cysts, tumors or abscess in the abdomen or liver.
- It is also commonly used to evaluate the structure of the thyroid gland in the neck.
- During pregnancy, an ultrasound can be used to evaluate the size, gender, movement, and position of the growing baby.

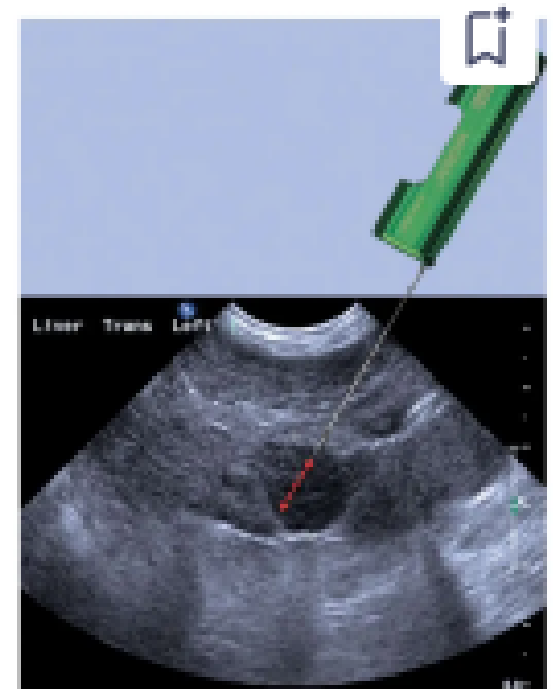
- One of the **most significant advantages** of US scans is **their ability to show movement in real-time**, unlike other more stagnant scans. Additionally, ultrasound is a safer imaging technique because there is no risk of ionizing radiation.

#### Advantages

- Non ionizing
- Cheap
- Portable
- Dynamic imaging
- Very good for cystic structures
- Biopsy, injection, aspiration

#### Disadvantages

- Highly operator dependent
- Only for superficial structures (cannot penetrate cortical bone)
- Limited field of view
- Poor resolution comparatively



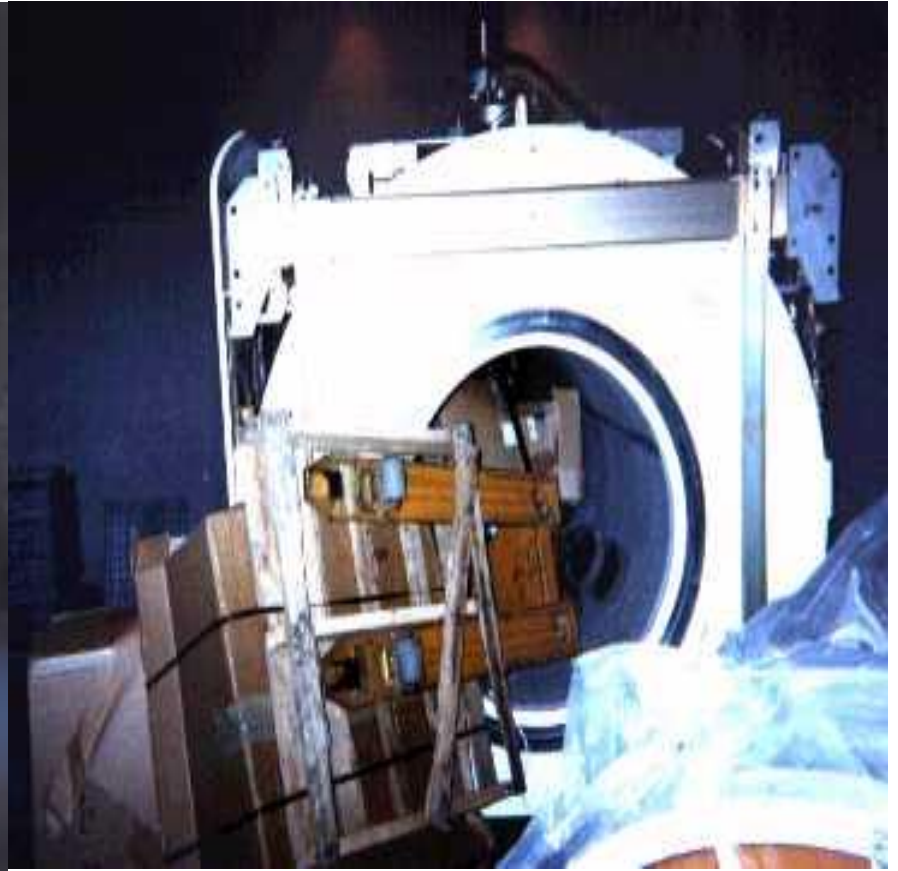
## Computed tomography (CT) scan

- Another commonly used medical imaging type is computed tomography (CT) scans **which combine** a more straightforward imaging technique, **X-rays**, with computer processing to image multiple body parts, providing a visual of bones, blood vessels, and organs.
- In traditional X-rays, energy beams are targeted at a specific body part and captured on a plate behind the body part for imaging. However, **CT scans move X-ray beams around the body to extract similar information and depict an image of the intended body part.**
- Unlike MRIs, **CT scans** require **patients to be exposed to some level of ionizing radiation.** Although the exposure is usually limited, it carries a **risk of DNA damage, which can result in other issues, including cancers.**
- Advantages: Rapid imaging, detailed visualization, multiplanar imaging, and simultaneous assessment.

## Computed Tomography (CT)

- During [a CT scan](#), the patient lies down on a table that slowly slides through the gantry, which is the circular part of the CT machine.
- Through the gantry, beams of x-rays are aimed at the body.
- The beams are rotated axially as the scan moves down the patient's body, and signals are sent to the machine's computer.
- This generates cross-sectional images called tomographic images.
- These images can be put together to form a 3D image of the patient's body, creating a more detailed view than normal x-rays.

# Magnetic Resonance Imaging (MRI)



## Magnetic Resonance Imaging (MRI)

- Magnetic resonance imaging (MRI) is one of the most recognized types of medical imaging. In this technique, the MRI machine — which is cylindrical — emits a magnetic field and computer-generated radio waves to craft detailed images of organs and tissues.

or

- It is a medical imaging technique used in radiology to form pictures of the anatomy and the physiological processes inside the body.
- It is a technique used to study soft tissue, such as the brain, spinal cord and muscles.
- It is used to reveal the anatomical structure of such tissues, as well as to monitor their functional characteristics or metabolism.
- MRI scanners use strong magnetic fields, magnetic field gradients, and radio waves to generate images of the organs in the body.

- The MRI suite can be a very dangerous place if strict precautions are not observed.
- Metal objects can become dangerous projectiles if they are taken into the scan room.
- For example, paperclips, pens, keys, scissors, hemostats, stethoscopes and any other small objects can be pulled out of pockets and off the body without warning, at which point they fly toward the opening of the magnet (where the patient is placed) at very high speeds, posing a threat to everyone in the room.
- Credit cards, bank cards and anything else with magnetic encoding will be erased by most MRI systems.

## Advantages:

- One of the primary benefits of MRIs is that they do not involve any radiation exposure, making it generally safe for patients.
- The additional benefits of MRI include soft tissue contrast, multiplanar imaging, and the ability to assess tissue function, blood flow, and more through functional MRI techniques.

## Disadvantages:

- Patients with metal implants, including pacemakers, cochlear implants and artificial joints cannot be imaged using MRIs because of the strong magnetic fields.
- Additional MRI limitations include cost and availability, longer scan times, and limited applications in acute trauma settings. Additionally, these machines require many resources to maintain.

## **Important question:**

1. What is medical imaging techniques?
2. Types of imaging techniques.
3. Write short notes on CT Scan.
4. Difference between CT Scan and MRI
5. Write application of X-ray.